

Python

The Programming Language: Origins, History & Uses

Overview

Python is a high-level, interpreted, general-purpose programming language. It's known for its readability and clean, simple syntax, which makes it one of the best languages for beginners to learn programming from scratch — while remaining powerful enough to build massive, complex systems used by major companies like Google, Netflix, Instagram, and NASA.

It was first released in 1991, is maintained today by the Python Software Foundation (PSF), is distributed under the open-source PSF License, and its files use the .py extension. As of early 2026, the latest major release is Python 3.13.

Who Created It?

Python was designed by the Dutch programmer **Guido van Rossum**. He began working on it in late 1989 at the CWI research institute in the Netherlands, during a Christmas holiday, as a personal project to keep himself occupied. Van Rossum wanted to create a successor to the ABC language that combined simplicity with power, addressing some of the flaws he'd noticed in it.

As for the name, it isn't inspired by the snake, as many assume, but by the famous British comedy series **Monty Python's Flying Circus**, of which van Rossum was a fan. Although the language's logo shows two intertwined snakes, van Rossum chose the name because he wanted something short, unique, and slightly mysterious.

He was later nicknamed the "**Benevolent Dictator For Life**" (**BDFL**) by the programming community, in recognition of his leading role in guiding the language's development for decades — before formally stepping down from that role in 2018, when decision-making passed to an elected Steering Council.

Key Historical Milestones

1989 — Guido van Rossum begins working on Python as a personal project. **1991** — The first official release, Python 0.9.0, comes out. **1994** — Python 1.0 is officially released, including functional programming tools. **2000** — Python 2.0 is released, introducing a cyclic

garbage collector.

2008 — Python 3.0 is released, a major overhaul that is not backward-compatible with Python 2. **2020** — Official support for Python 2 ends on January 1. **2022** — Python 3.11 is released with major performance improvements. **2023-2025** — Python 3.12 and 3.13 continue this trend, adding further performance gains and free-threading support.

Key Characteristics

Python's syntax is readable and closely resembles natural English, reducing complexity for newcomers. It is interpreted, meaning there's no need to compile code before running it, which speeds up development and testing. It is also dynamically typed, so there's no need to declare a variable's type in advance.

Python is multi-paradigm, supporting object-oriented, functional, and procedural programming all at once. It ships with a huge standard library — often described as "batteries included" — and benefits from a massive open-source community, with thousands of ready-made packages available through PyPI. It's also highly extensible, and can be combined with languages like C and C++ to boost performance where needed.

Main Areas of Use

Python is widely used in **AI and machine learning** (TensorFlow, PyTorch, scikit-learn), **web development** on the backend (Django, Flask, FastAPI), and **data science and analysis** (Pandas, NumPy, Matplotlib). It's also a popular choice for **automation and scripting**, handling repetitive tasks and system administration.

Beyond that, Python appears in **game development** through libraries like Pygame, in **scientific computing** for physics and astronomy research via tools like SciPy, and in **desktop applications** built with frameworks such as PyQt, Tkinter, and Kivy.

A Simple Python Code Example

```
print("Hello, World!")

def greet(name):
    return f"Hello, {name}!"

print(greet("Python"))
```

Fun Facts

Python isn't named after the snake, even though its logo features two of them. It is used in NASA space missions and in controlling robotic vehicles. The very first version of Python

didn't even have lambda functions or map/filter. The Python Software Foundation (PSF) is a non-profit that officially oversees the language's development, and Python has consistently been one of the most in-demand languages in the tech job market for years.

This report was prepared by Claude (Anthropic) as a sample PDF.